Yes. The client-side FreeForm-style architecture (natural-language search over video with appearance embeddings + soft biometrics) is highly relevant to a bank's organized crime prevention, physical security, and disaster response team. Below is a practical mapping of use cases, insights from leading companies and research, and an updated architecture that supports gender, age, size/shape, gait proxies, individual re-identification, and explicit absence queries — all runnable on GitHub Pages with open pretrained models.

Priority Use Cases for Your Team

1. Physical Security & Incident Investigation (Branches, ATMs, Vaults, Cash Areas)

- Rapid forensic search after robbery, assault, or suspicious activity using witness-style descriptions ("man in dark jacket, mid-30s, carrying black backpack, slight limp").
- Cross-camera and cross-branch correlation of the same individual during recon or multi-location activity.
- Restricted-zone / after-hours review: "Was anyone matching this description near the vault / cash room between 22:00–06:00?" or the inverse — confirm absence.
- ATM loitering, skimming, or jackpotting investigation with clothing + body-shape filtering.
- Object/person left-behind or unattended bag correlation with nearby people.

2. Organized Crime Prevention

- Linking individuals across separate incidents or branches when faces are obscured (masks, hats, distance) using clothing, gender, approximate age, height/build proxies, and gait style.
- Building temporary appearance profiles of suspected money mules or casing teams without maintaining a permanent facial biometric gallery.
- Multi-event pattern detection: same soft-biometric signature appearing at multiple ATMs or branches in a short window.
- Absence evidence: "Confirm this described person was **not** present at Branch X during the time of the disputed transaction."

3. Disaster Response / Business Continuity / Evacuation

- Post-incident staff accountability and missing-person search across the camera network using descriptions or soft biometrics.
- Evacuation verification: locate specific critical personnel or confirm they have left high-risk zones.

- Temporary facility / shelter headcount support and rapid search when systems are degraded.
- After-action review of movement during lockdown or emergency.

These map closely to capabilities already marketed by commercial platforms (attribute/people search, cross-camera tracking, watchlist alerts, behavioral analytics) but can be prototyped privately and offline with the open architecture.

Insights from Best-in-Class Companies, Labs & Guidance

- **BriefCam (now Milestone)** — Strong attribute search, people search, Review/Respond/Research modules. Widely used for rapid investigation in finance and public safety. Emphasizes filtering by clothing, direction, dwell time, and appearance.
- **Verkada, Spot AI, Kognition, March Networks, Axis, Genetec** — Banking-focused deployments covering ATM fraud, restricted-zone alerts, multi-branch correlation, loitering, and forensic search. Many stress camera-agnostic or hybrid edge approaches and privacy-aware design.
- **Academic / public-safety systems** — Multi-camera person re-identification for building evacuation and missing-person search (e.g., city-scale AI CCTV search systems, attention-aided Re-ID for evacuation tracking). Soft biometrics and appearance matching are standard when faces are unreliable.
- **Financial-sector context** — FS-ISAC-style threat sharing, NIST physical security guidance (PE-6 monitoring, video surveillance controls), and bank-specific analytics (ATM tampering, vault access, queue/occupancy, after-hours intrusion). Privacy and auditability are non-negotiable; many solutions deliberately avoid always-on facial recognition in public customer areas and prefer appearance + behavior.

Commercial systems typically combine detection, short-term tracking, attribute extraction, and searchable embeddings. Your open architecture can replicate the core search + soft-biometric layer for offline or pilot use while remaining fully under bank control.

Supporting Gender, Age, Size/Shape, Gait Proxies, Re-ID & Absence

All remain feasible client-side:

- **Gender & Age** — ONNX age-gender models (e.g., `onnx-community/age-gender-prediction-ONNX`) or face-api.js equivalents runnable via Transformers.js / ONNX Runtime Web.

- **Size / Shape / Proportions** — MediaPipe Pose Landmarker (33 keypoints) → normalized ratios (shoulder-hip, limb lengths, height proxy relative to frame, silhouette aspect).
- **Gait / Movement style** — Pose sequences over time → simple features (cadence proxy from ankle trajectories, left-right asymmetry, joint-angle patterns). Useful as a weak additional signal for short-term re-ID; not full clinical gait recognition.
- **Individual Re-ID** — Primary: CLIP appearance embedding. Secondary: soft-biometric vector (gender + age bin + body metrics + dominant colors). Tertiary: short pose-trajectory similarity when tracks exist. Weighted fusion or filter-then-rank.
- **Absence ("not in the video")** — Dual mode: (1) high-threshold "no confident match"; (2) attribute-coverage check (critical descriptors such as "bright yellow vest" or "red backpack" never appear above threshold). Always surface near-misses and rationale for human review.

These signals are especially valuable when faces are occluded, at distance, or when policy limits facial biometric use.

Refined FreeForm Architecture (GitHub Pages / Fully Client-Side)

Stack (all open / browser-ready)

- Detection: YOLO (n/s variants) via onnxruntime-web (WebGPU preferred).
- Pose & body metrics: MediaPipe Pose Landmarker (Web).
- Appearance + semantic search: Xenova CLIP (or MobileCLIP) via Transformers.js.
- Gender/Age: ONNX age-gender models.
- Storage: IndexedDB (embeddings + soft-bio vector + metadata + thumbnails).
- Search: CLIP text embedding + multi-signal ranking + absence logic.
- Optional: simple track formation (IoU + embedding + pose consistency).

Pipeline

1. User loads branch/ATM/incident videos (offline files).
2. Sample frames → detect persons → extract CLIP embedding + pose metrics + gender/age + colors.
3. Form short tracks where possible; store rich records.
4. Query (natural language or reference crop) → ranked results or absence report with confidence and near-misses.

5. Exportable index for sharing within the security team under controlled conditions.

Bank-specific enhancements to prioritize

- Strict audit logging of every search (who, what query, when, results viewed).
- Human-in-the-loop confirmation before any action.
- Configurable privacy modes (disable face crops, limit retention of embeddings).
- Multi-video / multi-location tagging so results are labeled by branch/ATM/camera.
- Simple watchlist of appearance profiles (temporary, policy-controlled).
- Export of evidence packages (clips + timestamps + similarity scores) for investigations or law-enforcement handoff.

Realistic scope for GH Pages
Excellent for offline forensic review, training, tabletop exercises, and rapid prototyping of search workflows. Not a replacement for production VMS with live multi-branch alerting, but a powerful, private, zero-cost analysis layer that security analysts can run locally or host internally.

Suggested Next Steps for Your Team

1. Pilot on historical incident videos (robbery, ATM events, after-hours alerts) to validate search quality with real descriptions.
2. Define soft-biometric fields and ranking weights that match your typical witness statements and camera quality.
3. Add absence-mode reporting templates suitable for internal reports or regulatory evidence.
4. Evaluate whether a later hybrid (static GH Pages frontend + internal GPU worker for heavier indexing) is needed once the workflow proves valuable.
5. Align with legal/compliance on data retention, purpose limitation, and whether temporary appearance profiles are treated as biometric under applicable law.

This architecture directly supports the high-value investigative and accountability workflows of organized crime prevention, physical security, and disaster response while remaining fully under your control and implementable with current open models.

I can next produce a concrete data schema, ranking formula, prioritized feature backlog, or a starter component outline tailored to branch/ATM video if that would be useful.